

The empirical recurrence for OEIS A306050, recovered from its tail

Adrian Perez Fontelles
Independent researcher

8 September 2026

Abstract

OEIS A306050 records “Empirical recurrence of order 85” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 141$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 5$, so the recurrence is proved for every $n \geq 90$. Its last coefficient is $c_{85} = -13312 \neq 0$, so no further tail satisfies anything shorter; any order-85 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A306050 is “Number of $n \times 5$ 0..1 arrays with every element unequal to 0, 1, 2, 3, 6 or 7 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

16, 120, 439, 1962, 12349, 66149, 321707, 1803281, ...

The entry states:

Empirical recurrence of order 85 (see link above)

As of the “Last modified” line on the live entry (Jun 18 2018 07:37:58, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 141$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 141$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 85: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 5$ is the first whose minimal order is exactly the stated 85.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 282$ exact terms from $n = 5$ on, it returns order 85 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{85} c_i a(n-i),$$

c_1	10	c_{23}	455083365	c_{45}	−9345183343	c_{67}	501033992
c_2	−32	c_{24}	−125550910	c_{46}	282874374	c_{68}	−27045081
c_3	122	c_{25}	−330903430	c_{47}	−173701928	c_{69}	−51322581
c_4	−760	c_{26}	−1008946154	c_{48}	9124975953	c_{70}	−171497477
c_5	1947	c_{27}	561739805	c_{49}	−991106258	c_{71}	−59284269
c_6	−3455	c_{28}	545096553	c_{50}	−2351382133	c_{72}	76631684
c_7	18796	c_{29}	1650161353	c_{51}	−5852210116	c_{73}	23331908
c_8	−46891	c_{30}	−1207469084	c_{52}	447731984	c_{74}	32001769
c_9	45279	c_{31}	−307750224	c_{53}	3896701167	c_{75}	−37276262
c_{10}	−232307	c_{32}	−2085170871	c_{54}	2157583166	c_{76}	4787694
c_{11}	627033	c_{33}	1453142888	c_{55}	675828923	c_{77}	−7812422
c_{12}	−312338	c_{34}	−400093288	c_{56}	−3634028523	c_{78}	9243308
c_{13}	1634969	c_{35}	1794002049	c_{57}	−174988874	c_{79}	−2444592
c_{14}	−5450640	c_{36}	−639588071	c_{58}	−1526614792	c_{80}	791856
c_{15}	986778	c_{37}	1266813343	c_{59}	2524527193	c_{81}	−1063648
c_{16}	−5948975	c_{38}	−296624273	c_{60}	−208914537	c_{82}	350016
c_{17}	33094179	c_{39}	−2003854198	c_{61}	1506754940	c_{83}	−13184
c_{18}	−256518	c_{40}	−1784594877	c_{62}	−1407323843	c_{84}	32768
c_{19}	1067384	c_{41}	−1443709189	c_{63}	186788519	c_{85}	−13312
c_{20}	−145134726	c_{42}	6168181547	c_{64}	−983621040		
c_{21}	6700257	c_{43}	1129187541	c_{65}	489223500		
c_{22}	84130679	c_{44}	1790187730	c_{66}	−84608894		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 90$, and it is the recurrence OEIS A306050 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 5$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{85} - c_1x^{85-1} - \dots - c_{85}$. Its constant term is $-c_{85} = 13312$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 85 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 85, so $\deg q = 85$, and it is monic. It holds from some index t on. From $\max(t, 5)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 85, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 20 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 85, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -13312 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-85 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A306050.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.